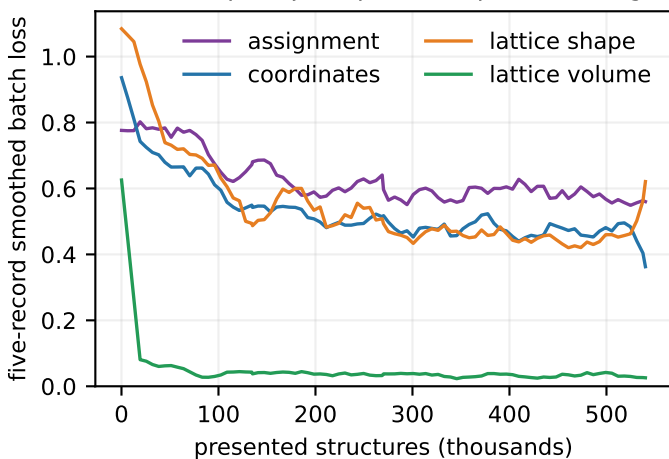
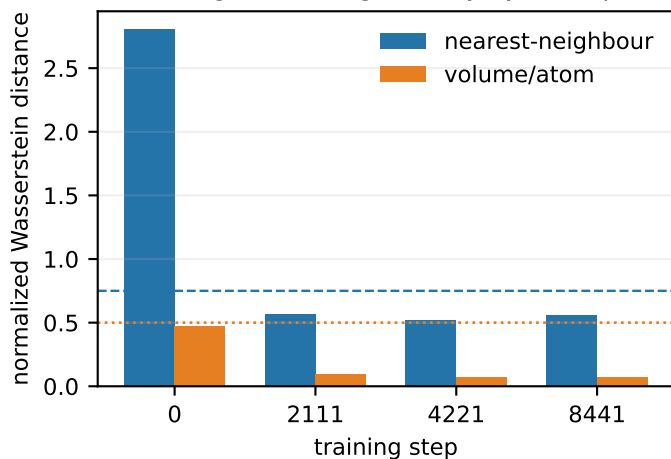


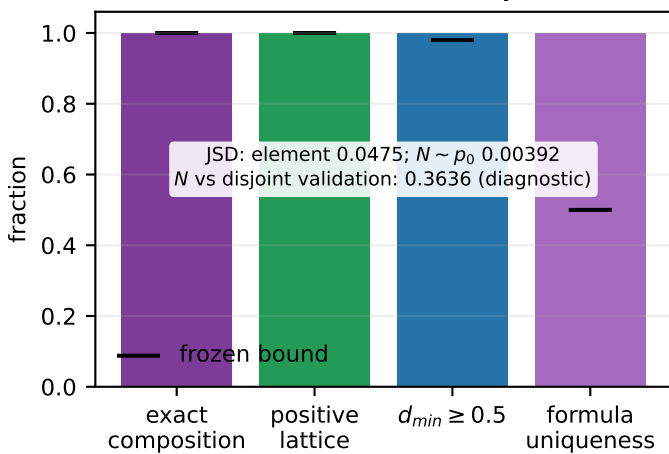
(a) One-pass joint product-space training



(b) Free-generation geometry by checkpoint



(c) Final discrete and validity closure



(d) Final local-packing quantiles

